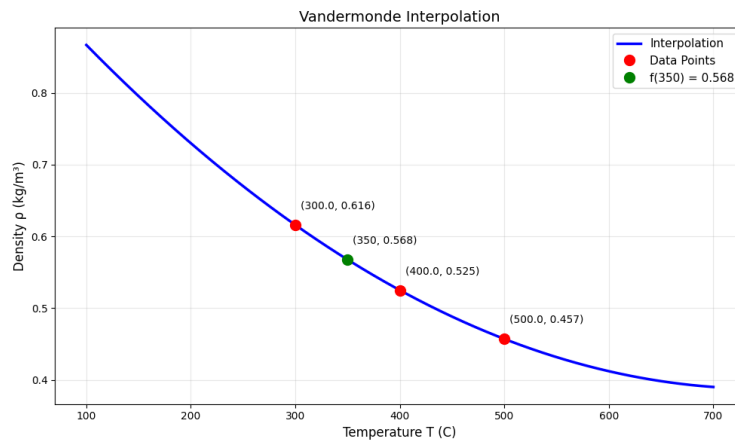


ECEN4293 Homework 3

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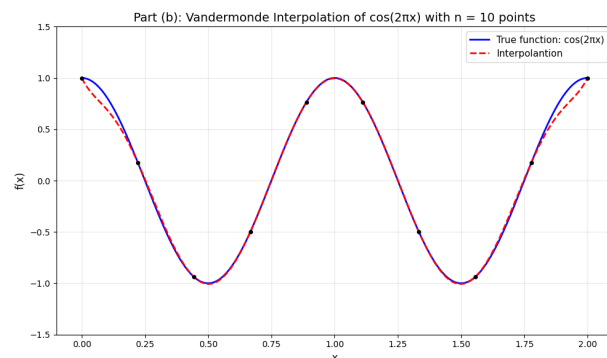
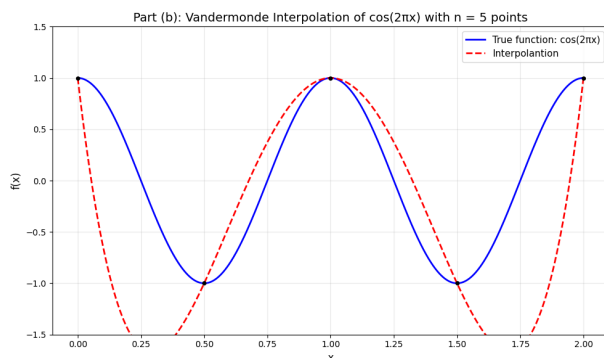
Problem 1 Vandermonde Interpolation

- (a) Write a Python function that accepts an array of ordered pairs and computes the Vandermonde interpolating polynomial. Your function should return a numpy Polynomial object. Test your function by replicating the 3-point density interpolation plot from lecture.



- (b) Use your function to interpolate $f(x) = \cos(2\pi x)$ with n linearly spaced points on the interval $\in [0, 2]$. Repeat this with $n = 5, n = 10, \dots, n = 95, n = 100$ and make a plot for each value of n . What do you observe? Include at least one plot in your write-up to support your observations.

With only 5 points ($n = 5$), the Vandermonde interpolation fails to accurately capture the frequency of the function $f(x) = \cos(2\pi x)$. However, increasing the sample size to 10 points yields an approximation that is visually indistinguishable from the true function, a level of accuracy that is maintained for datasets where $n > 10$.

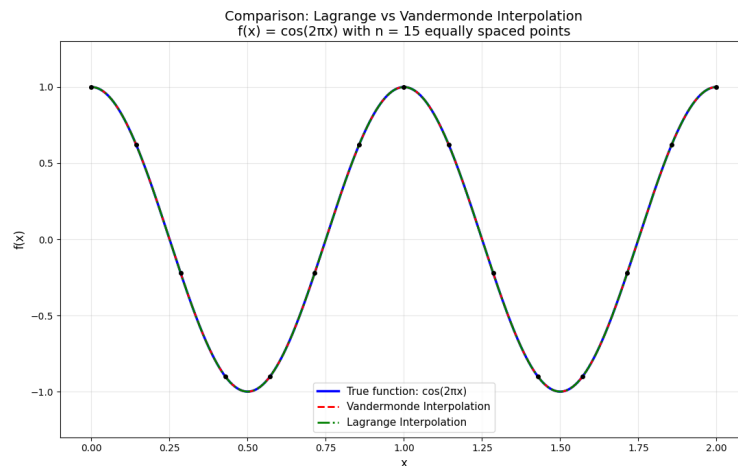


Problem 2 Lagrange Interpolation

- (a) Write a Python function that accepts an array of ordered pairs and computes the Lagrange interpolating polynomial.

Look at prob2.py

- (b) Test your function against your Vandermonde interpolation function from Problem 1 with an example of your choosing. Demonstrate the results with a plot.



Problem 3 Inverse Interpolation

Table 1: The refractive index of borosilicate glass at different wavelengths of light is presented in the table below.

Wavelength	Refractive Index
6563	1.50883
6439	1.50917
5890	1.51124
5338	1.51386
5086	1.51534
4861	1.51690
4340	1.52136
3988	1.52546

- (a) Find the Lagrange interpolating polynomial for the refractive index as a function of wavelength.

$$f(\lambda) = 29.06715812 - 0.03648098\lambda + 2.06323342 \times 10^{-5}\lambda^2 - 6.45218924 \times 10^{-9}\lambda^3 + 1.20420368 \times 10^{-12}\lambda^4 - 1.341004 \times 10^{-16}\lambda^5 + 8.24979339 \times 10^{-21}\lambda^6 - 2.16289805 \times 10^{-25}\lambda^7$$

- (b) Use a root-finding method of your choice with your interpolation to find the wavelength where the refractive index is 1.520.

Using the Newton-Raphson method with the interpolated function the wavelength was calculated to be 4485.35 where refractive index is 1.520.

